



Automated detection of incidental abdominal aortic aneurysms on computed tomography

Devina Chatterjee¹ · Thomas C. Shen¹ · Pritam Mukherjee¹ · Sungwon Lee¹ · John W. Garrett² · Nicholas Zacharias² · Perry J. Pickhardt² · Ronald M. Summers^{1,3} 

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Abstract

Purpose To detect and assess abdominal aortic aneurysms (AAAs) on CT in a large asymptomatic adult patient population using fully-automated deep learning software.

Materials and methods The abdominal aorta was segmented using a fully-automated deep learning model trained on 66 manually-segmented abdominal CT scans from two datasets. The axial diameters of the segmented aorta were extracted to detect the presence of AAAs—maximum axial aortic diameter greater than 3 cm were labeled as AAA positive. The trained system was then externally-validated on CT colonography scans of 9172 asymptomatic outpatients (mean age, 57 years) referred for colorectal cancer screening. Using a previously-validated automated calcified atherosclerotic plaque detector, we correlated abdominal aortic Agatston and volume scores with the presence of AAA.

Results The deep learning software detected AAA on the external validation dataset with a sensitivity, specificity, and AUC of 96%, (95% CI 89%, 100%), 96% (96%, 97%), and 99% (98%, 99%) respectively. The Agatston and volume scores of reported AAA-positive cases were statistically significantly greater than those of reported AAA-negative cases ($p < 0.0001$). Using plaque alone as a AAA detector, at a threshold Agatston score of 2871, the sensitivity and specificity were 84% (73%, 94%) and 87% (86%, 87%), respectively.

Conclusion Fully-automated detection and assessment of AAA on CT is feasible and accurate. There was a strong statistical association between the presence of AAA and the quantity of abdominal aortic calcified atherosclerotic plaque.

Keywords Abdominal aorta · Atherosclerotic calcification · Agatston score · Deep learning · CT

Abbreviations

AAAs	Abdominal aortic aneurysms
BTCV	Beyond the cranial vault
ROC	Receiver operating characteristic
AUC	Area under curve
DSC	Dice similarity coefficient

Perry J. Pickhardt and Ronald M. Summers are co-senior authors.

✉ Ronald M. Summers
rms@nih.gov

¹ Radiology and Imaging Sciences, Clinical Center, National Institutes of Health, Bethesda, MD 20892-1182, USA

² Department of Radiology, University of Wisconsin School of Medicine and Public Health, Madison, WI 53726, USA

³ Imaging Biomarkers and Computer-Aided Diagnosis Laboratory, Radiology and Imaging Sciences, National Institutes of Health Clinical Center, Bldg. 10 Room 1C224D MSC 1182, Bethesda, MD 20892-1182, USA

Introduction

An abdominal aortic aneurysm (AAA) is defined as a permanent localized dilatation of the aorta [1], and a diameter threshold of 3 cm is commonly used for the reporting of AAAs [2]. Untreated aneurysms can grow and eventually rupture, with mortality rates of 90% [3]. AAA can be detected in both contrast-enhanced and non-contrast-enhanced abdominal computed tomography (CT) exams but many go undetected [3, 4]. In one study, 35% of all AAAs and 48% of AAAs between 3.0 and 3.9 cm in diameter were not reported [4].

Since aneurysm surveillance programs have been shown to be beneficial, under-reporting and delayed diagnosis of AAA should be avoided [4]. Part of the challenge in implementing consistent detection practices is that aneurysm size measurements tend to have high inter-reader variability [5].

Automated methods have been shown to reduce inter-reader variability in a variety of imaging-based diagnoses

[6]. Examples include automated classification of prostate cancer severity and detection of brain metastases [6]. In addition to reducing variability, automated methods may reduce the burden of manual measurements; even for trained readers, manual determinations of the maximum aortic diameter can be time-consuming.

Previous literature on automated AAA detection has shown high accuracy [7]. However, previous works lacked testing in large external datasets that represent more general patient populations. Additionally, while the presence of atherosclerosis or aortic plaque is a known risk factor for AAA [8, 9], relatively few studies have studied the association between calcium scores and aortic aneurysms, particularly focusing on the abdomen [10, 11].

In this paper, we trained and validated fully-automated deep learning software to detect AAAs in abdominal CT and studied the relationship between AAAs and calcified atherosclerotic plaque.

Methods

The Institutional Review Board and Office of Human Research Subjects Protection at the respective institutions approved the use of the patient data. The need for informed consent was waived.

Patient population

Two datasets were used for training the AAA detector and a third was used for validation. The first of the two training datasets, Cohort 1, was the Beyond the Cranial Vault (BTCV) dataset, a publicly-available dataset gathered for a MICCAI workshop challenge [12] (Table 1). The dataset contains segmentations for 9 major organs; only the abdominal aortic segmentations were used. All 47 non-IV contrast scans in BTCV were used. The slice thicknesses ranged from 2.5 to 5.0 mm. Scanner manufacturer, average age, and gender distributions for the BTCV database are not available. Manual assessment found 2 scans had AAA, with maximum axial aortic diameter > 3 cm.

The second training dataset, Cohort 2, was an internal one from the NIH Clinical Center. This dataset is composed of 19 abdominopelvic CT scans of 17 patients (two patients had both noncontrast and contrast enhanced scans) gathered by performing a query for mention of AAA in radiology reports (Table 1). The presence of AAA in these 19 scans was verified by author DC. Author DC is a postbaccalaureate research trainee who was trained and supervised by two senior radiologists (RMS and SL) with 29 and 13 years of experience, respectively. The scans were done for a variety of clinical indications. Ten of the scans were performed without and nine with IV contrast. The slice thickness and spacing for the scans was 1 mm.

The validation dataset, Cohort 3, comprised patients from a consecutive series of adults who underwent CT colonography at the University of Wisconsin Hospital and Clinics from 2004 to 2016, as has been previously described [13, 14]. The current analysis differs in that it describes AAAs in the dataset. These scans were performed with oral contrast but no intravenous contrast. Only the initial scan for each patient was included. Patients missing an initial supine scan were excluded.

The prospective reports for the scans were also collected and searched for mentions of AAA using fuzzy string matching. Essentially, we searched for occurrence of words and word fragments such as “aort”, “aorta”, “abdominal”, “ectasia”, “aneurysm”, “AAA”, “dilation” and “dilatation” in the reports and extracted sentences containing any of these words. The sentences were manually reviewed to determine the ground truth presence or absence of AAA in the scan. Additional sanity checks were also performed to ensure the correctness of the labels: for example, a commonly occurring phrase in this cohort was: “AAA Screening: No evidence of aortoiliac aneurysm”; using string matching, we searched for reports containing this phrase and checked that all corresponding reports were marked AAA negative. Ambiguous cases were reviewed by a radiologist to establish ground truth labels.

Next, the trained AAA-detector was run on the scans, and the following additional exclusion criteria were applied based on the output segmentation and subsequent manual

Table 1 Patient and CT scan information

Dataset	Cohort 1 (training)	Cohort 2 (training)	Cohort 3 (testing)
Number of patients (n)	Not available	17	9172
Number of scans (n)	47	19	9172
Scanner manufacturer	Not available	16 Siemens, 3 Toshiba	All GE Healthcare
Slice thickness (mm)	2.5–5.0	1.0	1.25
IV contrast (n)	0	10	0
AAA (n)	2	11	49
Age (yrs, mean $\pm$ std.)	Not available	68 $\pm$ 13	57 $\pm$ 8
Male patients (n)	Not available	15 (88%)	4054 (44.2%)

review: (a) full aorta not included in the scan, determined by an aorta segmentation spanning less than 5 slices, (b) severe artifacts or noise that resulted in failed segmentations, (c) serpentine or tortuous aorta, determined by the failure of the centerline fitting algorithm, which was configured to fail on highly curved vessels, (d) patients with endostents or evidence of prior surgical abdominal aortic aneurysm repair mentioned in the reports.

Data processing

The original abdominal aortic segmentations in Cohort 1 (BTCV) were expanded by author DC to include calcified plaque in the segmented region if the plaque was excluded by the original annotators. The segmentations were modified utilizing ITK-SNAP V3.6.0 [15].

The abdominal aortas in Cohort 2 were manually segmented by author DC and reviewed by two radiologists (RMS and. SL). These segmentations of aneurysmal abdominal aortas were used as training examples to ensure that the deep-learning based aorta segmenter would perform well on both normal and aneurysmal aortas.

The CT scans for Cohort 3 were originally in DICOM format but were converted to NifTi format. The scans were then resized to a slice spacing of 3 mm. Five scans were segmented by author DC utilizing ITK-SNAP, and used for validation of the aortic segmentation performance.

Manual measurement of AAAs

Following the usual clinical practice, maximum external (outer wall) oblique aortic diameters were manually measured in the axial plane by author DC on the reported AAA-positive patients in ITK-SNAP. No automatic centerline tool was utilized during the manual measurements. Axial, sagittal and coronal planes were utilized to identify the axial slice having the maximum aortic diameter. Author DC was blinded to any prior diameter measurements and independently measured the maximum external aortic diameters on the AAA-positive scans.

Automated AAA detection system

We created fully-automated software that segments the abdominal aorta between the aortic hiatus and bifurcation, uses ellipse-fitting on each axial slice of the segmentation to extract maximal aortic diameter for the given slice, and creates a maximal aortic diameter profile across axial slices for the determination of AAA status.

The aorta segmenter used a previously published modified 3D U-Net architecture [13]. The segmenter was trained using the 66 scans in the Cohorts 1 and 2 described earlier. The ellipse fitting consisted of two

steps. First, med2image [16] was used to convert individual axial slices of the predicted 3D aortic segmentation into 2D JPEG images. Next, ellipse fitting was performed on each 2D axial image using OpenCV [17] via the Python package “opencv-python” (version 4.8.0.76). Maximal diameter determinations were made on each axial slice using the major diameter of the ellipse. For any axial slice, if the extracted aortic diameter was greater than 3 cm, the scan was labeled positive for AAA.

Automated atherosclerotic plaque detection system

We used a previously-validated fully-automated deep learning calcified abdominal aortic atherosclerotic plaque detector to quantify the abdominal aortic calcified plaque burden [13]. The detector segments and measures all of the calcified plaque between the aortic hiatus and aortoiliac bifurcation. In order to reduce false positive plaque, a circular filter with radius 6 cm centered in each 2D CT slice is used, and all detections outside the circle are removed. The detector was trained using a modified 3D U-Net architecture [18] on low-dose PET-CT scans of patients with vasculitis and routine CT urography scans. It outputs the total abdominal aortic Agatston and volume scores for each CT scan [13]. These scores may contain contributions from calcification in neighboring and branch vessels such as the celiac and superior mesenteric arteries. To test the robustness of our conclusions, we also computed Agatston and volume scores by removing all plaque detections outside the abdominal aorta using the aorta segmentations obtained from the aorta segmenter.

Performance evaluation and statistical analysis

To evaluate the performance of the abdominal aortic segmenter, we calculated the Dice Similarity Coefficient (DSC) on five manually segmented aortas in Cohort 3. The diameter error, defined as the absolute difference between the automated and manual maximum diameter determinations, was computed for the true positive AAA detections.

Next, we evaluated the sensitivity, specificity and positive predictive value of AAA detection using the maximum diameter threshold of 3 cm. We also constructed receiver operating characteristic curves (ROC) to analyze how the sensitivity and specificity varied with changing the maximum diameter threshold for assigning a AAA positive label. The “pROC” package in R was utilized for the ROC analysis.

A Student’s *t*-test assuming unequal variance was performed to compare the distributions of abdominal aortic Agatston and volume scores between the reported AAA-positive and AAA-negative population. The *t*-test was done with the “scipy” Python package [19]. ROC analysis was

also performed to test how well Agatston score predicts AAA. The Youden index was used to determine optimal operating points on the ROC curves.

Bootstrapping with 2000 iterations were used to compute 95% confidence intervals for the reported metrics including sensitivity, specificity and the area under the ROC curve (AUC).

Results

External validation cohort

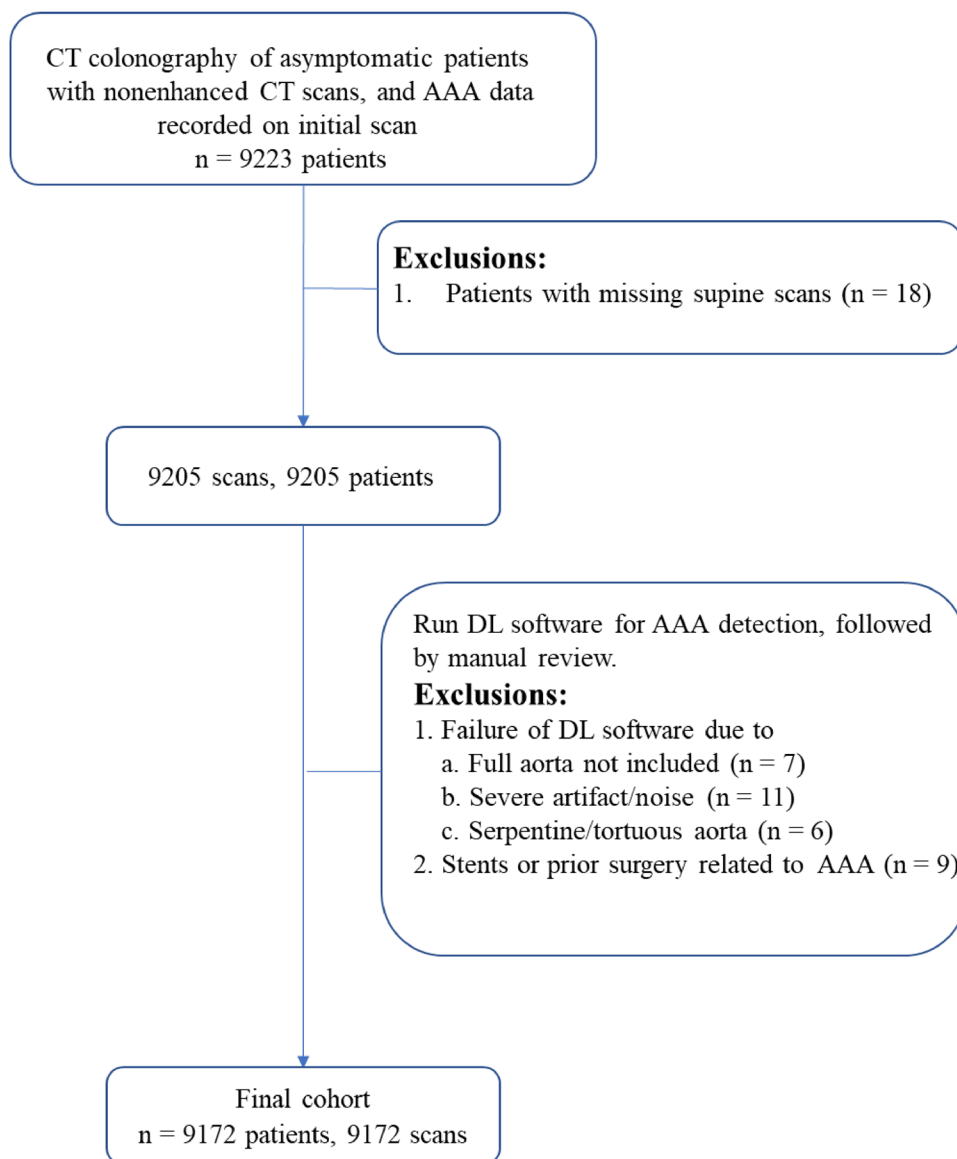
The external dataset, Cohort 3, consisted of 9223 unique patients who underwent CT colonography scans with oral but without intravenous contrast. Eighteen scans (18/9223,

0.002%) were excluded since their initial noncontrast supine scan was missing (Fig. 1). In 24 scans, the automated software was unable to segment or analyze the abdominal aorta due to severe artifacts or noise ($n = 11$), lack of a full abdominal aorta contained within the scan ($n = 7$), or serpentine or tortuous aortas ($n = 6$). An additional 9 patients were excluded since their reports indicated the presence of stents or prior surgery to repair AAAs. After exclusions, the final validation cohort consisted of 9172 scans (mean age, 57 ± 8 years; 4054 males).

Abdominal aortic segmentation

The software successfully produced a segmentation of the abdominal aorta in 99.7% (9181/9205) of scans. The

Fig. 1 Overview of Cohort 3, the validation cohort



mean DSC was 0.94 ± 0.02 on the five manually segmented abdominal aortas in the external validation dataset.

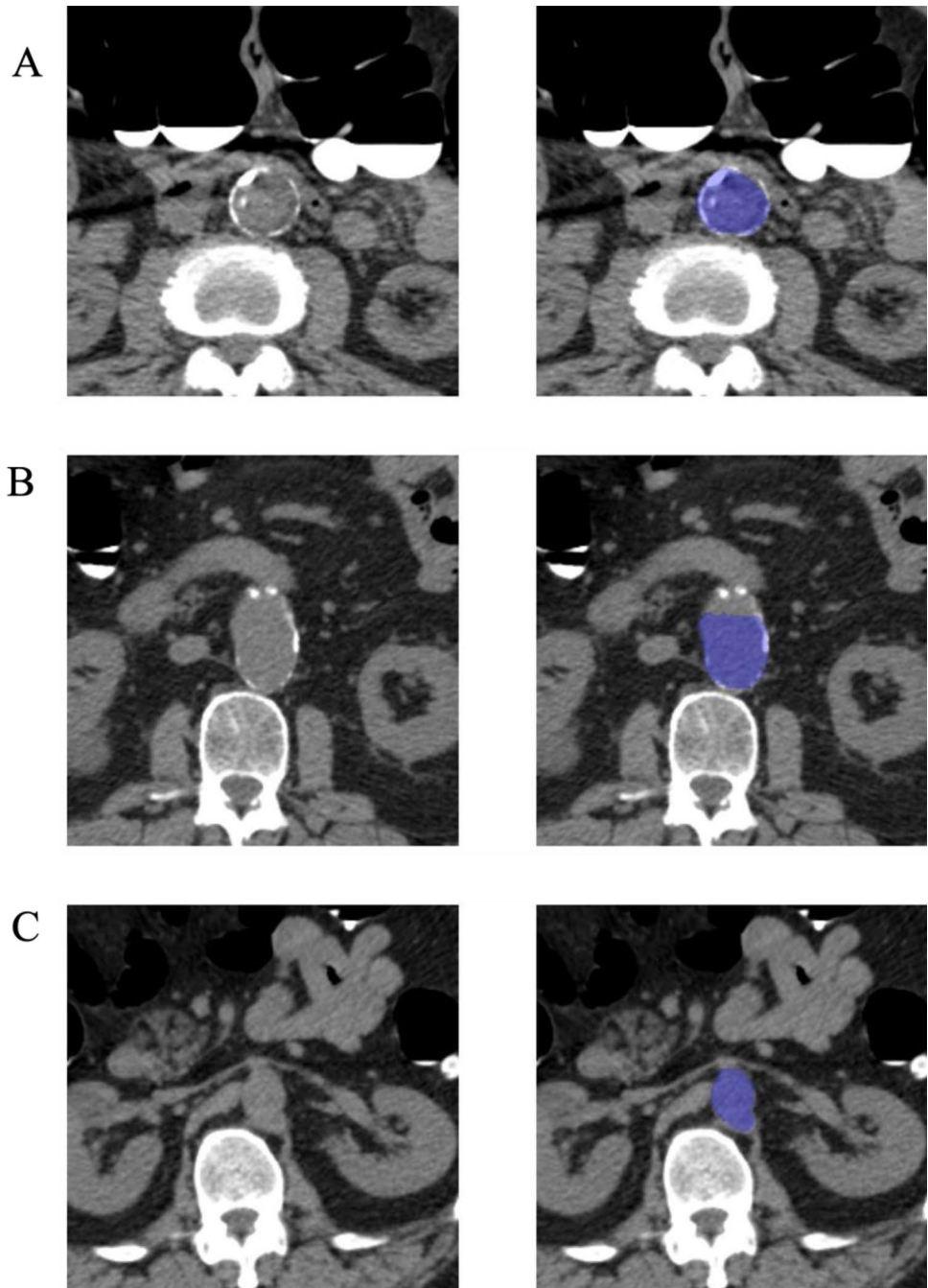
AAA detection in the external validation dataset

The prevalence of AAAs reported at prospective interpretation was 0.5% (49/9172). Figure 2 shows some examples of aortic segmentations for true positive and false positive AAA detections, while Fig. 3 shows the aortic diameter profile on a reported AAA-positive patient and a reported AAA-negative patient. At a threshold of 3 cm, the detector had a

sensitivity and specificity of 96% (47/49) [95% CI (89%, 100%)] and 96% (8796/9123) (96%, 97%), respectively. The positive predictive value was (47/374) 13% (9%, 16%). The ROC curve is shown in Fig. 4. The AUC was 99% (98%, 99%). At the Youden index, the maximum diameter threshold for detecting AAAs was 2.8 cm for which the sensitivity was 100% and the specificity was 93%.

For the 47 scans on which the AAA was correctly detected, the median and mean diameter errors were 2 and 5 mm, respectively. The standard deviation of the mean difference was 8 mm. The large difference between the mean

Fig. 2 Examples of aortic segmentation: **A** An accurate segmentation and true positive AAA detection for a 73 yr old male (maximum axial diameter 3.2 cm). **B** Undersegmented aorta for a true positive AAA detection (maximum axial diameter: 7.9 cm manual vs 4.2 cm automated) in a 73 yr old male. The undersegmentation and maximum automated measurement occurred on the slice shown, but the maximum manual diameter was on a different slice (not shown). **C** Oversegmented aorta for a false positive AAA detection in a 59 yr old male



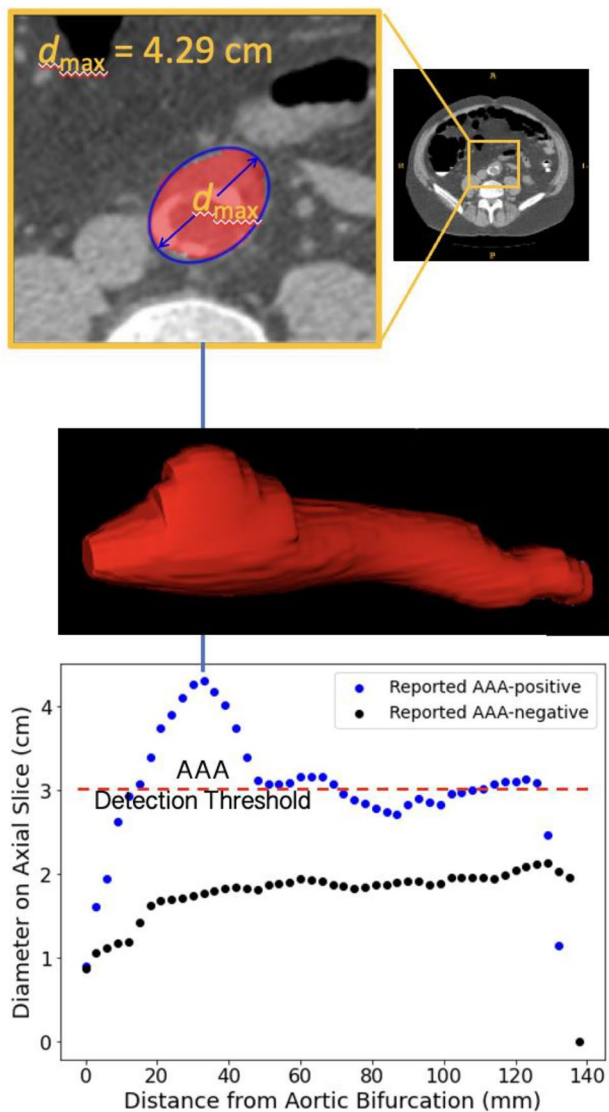


Fig. 3 Diameter profiles of an AAA patient (blue) and a non-AAA patient (black). Example images highlight the ellipse (blue) that is extracted from the segmentation model. The red-shaded region of the image is the aortic segmentation. The axial CT slice shown is from the scan that had a reported AAA; the AAA is 33 mm superior to the aortic bifurcation at the detected peak. The plot shows the length of the major axis of the ellipse fitted on each aortic segmentation slice as a function of position in the body. The diameter profile of the AAA-negative patient never exceeds 3 cm. In contrast, for the AAA-positive patient, there is a bulge that rises above the 3 cm threshold. The red 3D rendering shows the segmentation and AAA. On the horizontal axis, 0 mm corresponds to the axial slice at the aortic bifurcation; increasing values correspond to increasingly superior slices of the aorta. At the level of the aortic hiatus, 140 mm away from the aortic bifurcation, there are trailing data points for both patients. These data points correspond to the tapering of the segmentation at the superior end of the aorta

and median was a consequence of large errors (> 1 cm) in seven of the 47 true-positive AAA cases. In five of these cases, there were significant under-segmentations of the

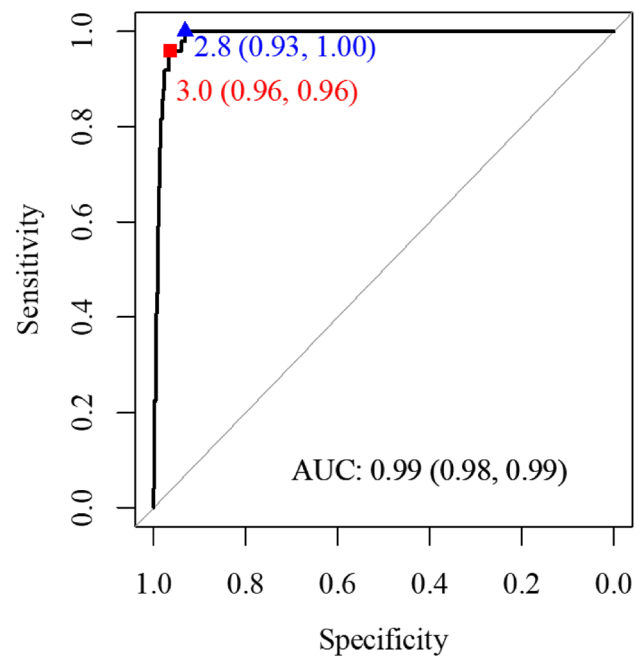


Fig. 4 ROC Curve for diameter thresholding. The AUC was 0.99 [95% CI (0.98, 0.99)]. The Youden-index-maximizing point of 2.8 cm is highlighted as the blue triangle (specificity: 0.93, sensitivity: 1.0), while the clinical standard of 3 cm used in this paper is highlighted as the red square (specificity: 0.96, sensitivity: 0.96)

large aneurysms (mean maximum diameter: 5.9 cm, STD: 1.4 cm, automated measurement: 3.6 cm, STD: 0.4 cm).

Manual review of 2 missed cases found that one of them was a large aneurysm > 6.4 cm in diameter (automated measurement: 2.8 cm) while the other was borderline with a manually measured maximum diameter of 3.0 cm (automated measurement: 2.8 cm). Of the 327 false positive scans, 25 scans were found on manual review to have significant noise and/or artifacts. The maximum diameter measurement for the 327 false positive cases had a median of 3.2 cm [IQR: (3.1, 3.4) cm].

Plaque association

The distributions of the Agatston and volume scores between AAA-positive and AAA-negative scans were statistically significantly different ($p < 0.001$). Figure 5 shows the distribution of calcified plaque scores in Cohort 3. The false-positive scans have higher Agatston and volume scores than the AAA-negative scans, but lower scores than AAA-positive scans. Figure 6 shows the ROC curve for the association between Agatston score and the presence of a AAA. The AUC was 0.86 [95% CI (81%, 91%)]. At the optimal operating point, with an Agatston score threshold of 579, determined by maximizing the Youden Index, sensitivity and specificity were 82% (71%, 92%) and 81% (80%, 82%),

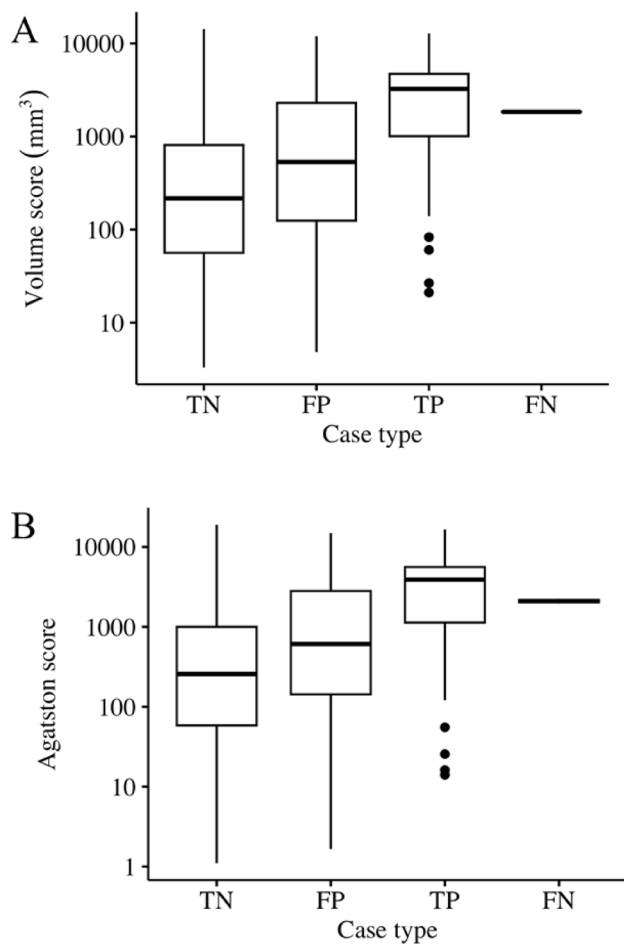


Fig. 5 Volume and Agatston score distributions based on the presence and detection of AAA. Here, *TN* True negatives, *FP* False positive detections, *TP* True positive detections, *FN* False Negative (missed) detections. **A** Volume scores distribution in Cohort 3 **B** Agatston scores distribution in Cohort 3

respectively. Corresponding results obtained by removing all plaques outside the aorta segmentations largely agree with the above results and are presented in Supplementary Figures 1 and 2.

Discussion

We developed fully-automated software that detects AAAs on abdominal CT and validated it in a large external cohort of asymptomatic adults. The detector performed well, reaching high sensitivity and specificity in external validation. We also found a strong association between AAAs and abdominal aortic calcified plaque.

The prevalence of AAA in our study was 0.5% (47/9172) using prospective readings. Khashram et al. found the AAA prevalence to be 4.8% in a patient population of 4644 patients in New Zealand who were undergoing

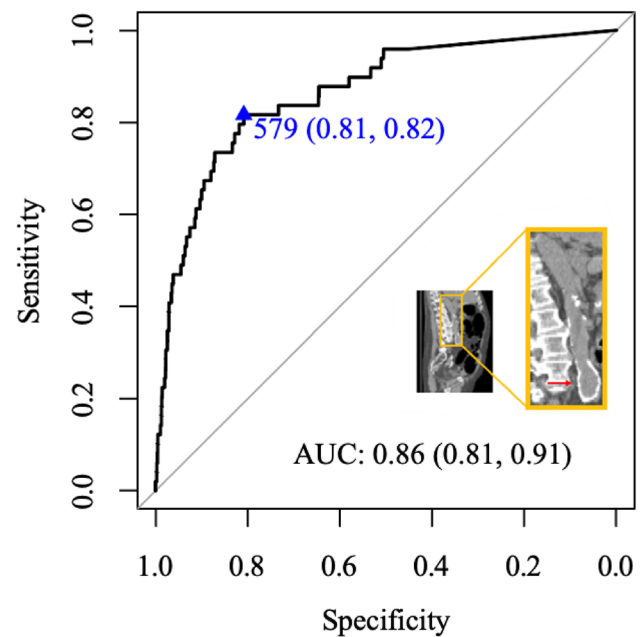


Fig. 6 ROC curve using Agatston score alone (without the aid of ellipse-fitting-derived diameter information) as a detection variable on the external validation dataset ($n = 9172$). The operating point (Agatston score) of 579 is highlighted as a blue triangle and is chosen by maximizing the Youden index. The AUC was 0.86 [95% CI (0.81, 0.91)]. The inset images show a sagittal view of the aorta on a reported 81-year-old male AAA-positive patient (red arrow indicates aneurysm), with an Agatston score of 11924

CT colonography for gastrointestinal symptoms [20]. The difference in prevalence may relate to the older patient population (median age of 79.6 years, versus 57 years in our study).

Two prior studies developed automated tools for AAA detection on CT angiography of preoperative patients with known AAA. Adam et al. utilized training data with 489 scans and an internal validation data set of 62 scans without utilizing any external data; the patient population was comprised of patients undergoing open or endovascular repairs [21]. The authors found a mean absolute diameter difference of 1.2 mm between the automated measurements and reference standard, comparable to that of seven expert annotators whose differences ranged from 1 to 2 mm. Brutti et al. analyzed the aorta and aortic thrombus on preoperative CT angiography scans of patients diagnosed with AAA as their primary pathology; their test set consisted of only 14 scans [7]. The authors found a mean absolute diameter difference of 3.2 mm. In comparison, we found a mean absolute diameter difference of 5 mm.

Our software also undersegmented a few cases of large aneurysms—this led to a large difference (> 1 cm) in the manual and automated measurements of maximum diameter in five true-positive cases and a missed AAA detection. This may be due to the paucity of training examples of

large aneurysms, as the maximum aortic diameter among the training examples was 4.4 cm.

Multi-planar reformatted views perpendicular to the aortic path have been advocated for measuring the diameter of a AAA [22]. In routine clinical practice, and for the reference standard used in this study, multi-planar reformatted views are not typically obtained; measurements are made from routine axial images. Since the intent of this tool is opportunistic screening rather than treatment planning for AAA repair, and to align with the method of the reference standard, the software made measurements on routine axial images.

A diameter threshold of 3 cm, as used in this study, is commonly used in radiology for the reporting of AAAs [2]. More recent evidence suggests that a smaller size threshold might be required to detect AAAs that subsequently ruptured [23]. However, a smaller threshold would also lead to many more AAA diagnoses and potentially unnecessary screening. If the reporting threshold were reduced, the automated tool may require revalidation.

To our knowledge, the quantitative association between abdominal aortic calcification and the presence of AAA has yet to be studied in a large dataset. The two pathologies have similar biomarkers: high fibrinogen and CRP, and low HDL [24]. A potential mechanism for the causative role of atherosclerosis in AAA development has been arterial remodeling [25]. Ward et al. hypothesized that the inflammatory reactions that occur from the aortic atherosclerotic plaque cause the breakdown of the extracellular matrix [25]. To compensate, the aortic wall may expand in response to the sheer stress alterations. While we found a strong association between AAA and calcified plaque, we did not investigate any causal relationship between the two. The underlying mechanisms, if any, will require further study.

The association between abdominal aortic calcification and AAA status appears so strong that abdominal aortic calcification alone (without the aid of aorta segmentation) can be considered an indicator for AAA status. On the external validation dataset, such a “calcification-only” AAA detector achieved an AUC of 0.86 [95% CI (0.81,0.91)] (Fig. 6), a sensitivity of 82%, and a specificity of 81%. Since calcified abdominal aortic plaque increases with age [26], future work will need to assess this observed relationship in patients in different age groups.

There are limitations to this study. The incidence of reported AAA in the external validation dataset was low, a possible indication of underreporting of borderline AAAs, a younger average patient age, and a generally healthy screening cohort. Also, the AAA detector was trained and validated primarily on noncontrast scans. Additional training and validation will be needed for CT scans performed with intravenous contrast. Finally, the final cohort was selected based on the output of the AAA detector and subsequent

manual review, instead of first excluding problematic scans with noise/artifacts, insufficient span of the aorta, and then running the detector on the clean cases. This was unavoidable since manually reviewing > 9000 scans for quality was infeasible; however, we manually reviewed each case excluded due to scan quality issues (24/9205, < 0.3%), and verified there were no AAA positive cases among them.

In summary, we presented and externally validated on a large patient dataset a fully-automated deep learning-based AAA detector for CT scans, and related the presence of AAAs to the severity of calcified plaque. The AAA detector demonstrated high sensitivity and specificity on the external validation dataset. This AAA detector may be useful in the setting of routine clinical interpretation of abdominal CT scans.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00261-023-04119-1>.

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Declarations

Conflict of interest There are no competing interests or conflicts of interest for any author. The authors wish to disclose the following non-competing interests: PJP: advisor to Bracco, GE Healthcare, and Nano-X. RMS receives royalties from iCAD, ScanMed, PingAn, Philips, Translation Holdings, MGB. His lab received research funding through a Cooperative Research and Development Agreement with PingAn.

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